

Pandas Practice

iris_dataset · 150 rows · 5 columns

 SYNTAX CHEATSHEET

First 5 rows	<code>df.head()</code>	Count per category	<code>df['col'].value_counts()</code>
Shape (rows, cols)	<code>df.shape</code>	Unique values	<code>df['col'].unique()</code>
Data types + nulls	<code>df.info()</code>	How many unique	<code>df['col'].nunique()</code>
Summary statistics	<code>df.describe()</code>	Filter rows	<code>df[df['col'] > value]</code>
Mean of column	<code>df['col'].mean()</code>	Filter by text	<code>df[df['col'] == 'text']</code>
Check missing values	<code>df.isnull().sum()</code>	Group + aggregate	<code>df.groupby('col')['col2'].mean()</code>
Sort values	<code>df.sort_values('col')</code>	Largest N values	<code>df.nlargest(n, 'col')</code>

► RUN THIS FIRST IN YOUR NOTEBOOK — DO NOT CHANGE IT

```
import pandas as pd
from sklearn.datasets import load_iris

iris_raw = load_iris()
df = pd.DataFrame(iris_raw.data, columns=['sepal_length', 'sepal_width',
    'petal_length', 'petal_width'])
df['species'] = iris_raw.target_names[iris_raw.target]

# Columns: sepal_length | sepal_width | petal_length | petal_width | species
```

QUESTIONS

Q1 FIRST LOOK

Display the first 5 rows of the dataset.

Q2 DATA INFO

Show data types and check if there are any missing values.

Q3 SPECIES COUNT

How many species are in the dataset? How many flowers per species?

Q4 SUMMARY STATISTICS

Show the summary statistics (mean, std, min, max) for all numeric columns.

Hint: One function does all of this at once.

Q5 GROUPBY

What is the average `petal_length` for each species?

Q6 FILTER

Filter and display only flowers where `petal_length` is greater than 5. How many rows did you get?

Q7 FILTER + GROUPBY

From the flowers with `petal_length > 5`, what is the average `sepal_width` per species?

Hint: Filter first, then groupby on the result.

Q8 ★ BONUS

Which species has the largest average `petal_width`? Show all species sorted from highest to lowest.